

TradePro Elite: A Real-Time Indian Stock Market Simulation Platform with AI-Powered Insights and Social Trading

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Abstract—Learning to trade in financial markets has traditionally required either risking real capital or settling for outdated, static simulations that bear little resemblance to actual market conditions. Neither approach serves the growing population of retail investors in India effectively.

This paper presents TradePro Elite, a comprehensive Indian stock market simulation platform that bridges this gap by combining live NSE/BSE market data, artificial intelligence, and community-driven learning into a single cohesive system. Real-time stock prices are streamed through a Google Sheets and Fyers API pipeline, giving users an authentic trading experience without any financial risk. An AI layer handles price movement prediction, news sentiment analysis, portfolio optimization, and automated risk assessment. A social trading module allows users to share strategies, follow top performers, and learn from the community.

The platform is built on a React and TypeScript frontend with WebSocket-driven live updates, backed by a Node.js and Express.js server, PostgreSQL with Prisma ORM for data persistence, and Redis for caching. Each user begins with a simulated balance of Rs.10,00,000 and has access to the full suite of trading tools including multiple order types, stop-loss, take-profit, and comprehensive portfolio analytics.

Testing confirms that the architecture delivers stable real-time performance and that the AI features produce actionable insights across a range of market conditions. TradePro Elite demonstrates that professional-grade trading education can be made accessible to everyone.

Index Terms—stock market simulation, paper trading, NSE/BSE, real-time market data, artificial intelligence, sentiment analysis, portfolio management, social trading, fintech education, Indian stock market

I. INTRODUCTION

India's retail investor base has grown at a pace that the country's financial education infrastructure has struggled to

keep up with. Millions of first-time investors enter the market each year, often with little preparation and no safe environment in which to develop their skills before putting real money at stake. The consequences are predictable: poorly timed decisions, avoidable losses, and a loss of confidence that pushes many people away from investing entirely.

The tools currently available to this audience fall into two unsatisfying categories. On one end are real brokerage platforms, which offer professional-grade features but no tolerance for the kind of trial and error that learning requires. On the other end are educational simulations, most of which rely on delayed or synthetic data and strip away the complexity that makes real trading genuinely instructive. Neither category addresses the full problem.

TradePro Elite is built to fill that space. It connects directly to live NSE and BSE market feeds through a Google Sheets and Fyers API pipeline, so every trade a user places reflects actual market conditions at that moment. An integrated AI layer provides price predictions, sentiment analysis drawn from financial news, and portfolio optimization recommendations. A social trading module lets users share strategies, publish their portfolio performance, and learn from a community of peers. All of this runs in a risk-free environment where mistakes cost nothing but remain as informative as they would be with real capital.

The system is designed as a modular, multi-layered architecture. The frontend, built with React 18 and TypeScript, handles everything the user sees and interacts with. The backend, developed on Node.js and Express.js, manages authentication, trading logic, and transaction processing. PostgreSQL with Prisma ORM stores persistent data, and Redis handles caching

to keep latency low under real-time load. WebSockets carry live price updates to the client the moment they arrive.

Contributions of this paper:

- A full-stack Indian stock market simulation platform that uses live NSE/BSE data via Google Sheets and Fyers API, providing an authentic trading experience without financial risk.
- An integrated AI layer covering price movement prediction, news sentiment analysis, portfolio optimization, and automated risk assessment.
- A social trading and community module that enables strategy sharing, leaderboards, and peer learning within the simulation environment.
- A modular backend architecture with real-time WebSocket updates, Redis caching, and a comprehensive PostgreSQL schema supporting portfolios, transactions, watchlists, alerts, and AI insights.

II. LITERATURE SURVEY

Prior work on stock market simulation and trading education spans a wide range, from conceptual studies on paper trading to practical platform implementations. The following survey examines the most relevant contributions and identifies the gaps that TradePro Elite addresses.

Sarawade et al. [1] developed a secure and modular real-time stock trading system intended for educational simulation. While the work demonstrates a sound approach to building trading infrastructure, the platform is limited in scalability and real-time responsiveness, relying primarily on static simulation environments without deep integration with live market feeds.

Chansarkar [2] examined the concept of paper trading as a method for risk-free learning of stock market operations. The study makes a strong case for simulation-based education but lacks integration of real-time data, automated feedback mechanisms, community collaboration, AI assistance, and advanced portfolio analytics.

Pagire et al. [3] built a mobile and web application for simplified stock market education. The system prioritises accessibility but falls short on trading realism, offering no advanced order execution, AI-driven insights, or live data synchronisation, which limits its effectiveness as a professional simulation tool.

Bhattacharya et al. [4] designed a virtual stock trading platform combined with structured educational tutorials. The platform does not integrate real-time data or AI support, and its educational content is predefined rather than adaptive to user progress.

Bouadjenek et al. [5] examined how social media sentiment influences stock market predictions. While insightful, the research remains theoretical and does not integrate findings into a practical trading simulation with real-time execution or feedback for traders.

Sharma et al. [6] analysed the impact of internet penetration on Indian online trading systems. The study does not include simulation or training components for beginners and offers no AI-based insights or predictive analysis.

Priyadarshini [7] assessed investor trust and behaviour towards online trading platforms through survey-based analysis. The research does not propose a technological solution and fails to address educational tools or simulation frameworks for skill development.

Maddodi and Kumar [8] reviewed AI and statistical methods used for predicting stock market movements, providing a strong theoretical foundation. However, the models reviewed are not deployed in practical learning simulations or interactive educational environments.

Trivedi and Bhatt [9] developed an educational platform for simulated stock trading. The study lacks real-time data usage, AI-based analytics, community features, and competitive learning elements.

Modi and Joshi [10] evaluated whether activities on social trading platforms could predict market movements through observational analysis. The work does not extend into trading education or simulation systems.

The NSDL Team [11] provided foundational educational material on stock selection. As a reference document rather than a system, it does not address simulation, real-time data, or platform development.

Alamelu and Indhumathi [12] studied technological factors contributing to online trading growth in India, focusing on infrastructural and economic aspects rather than user-oriented simulations, and lacking AI integration or experimental validation.

Desai and Joshi [13] created an AI-powered system for predictive trading and analytics. While technically capable, the setup is complex and not designed for paper trading environments or educational simulation.

Patel and Shah [14] designed a simulation-based trading platform with educational tutorials. The system does not include AI-driven analytics or real-time stock data, and its educational modules lack adaptability and community learning support.

Patel and Shah [15] explored innovative designs for next-generation trading platforms but did not address risk-free learning, educational simulation, or AI integration.

Patel and Chauhan [16] analysed fintech innovations improving trade digitisation through APIs, focusing on enterprise systems rather than educational or simulation purposes.

Shah and Patel [17] compiled a systematic survey of recent research trends in stock market prediction. As a review study, it lacks implementation or real-time validation.

Shah and Mehta [18] evaluated machine learning techniques for stock price forecasting without deploying or testing models in real-world trading or simulation environments.

Shah and Rawal [19] demonstrated paper trading as a tool to improve financial literacy at a small conceptual scale, without AI integration, analytics, or scalability.

Patel and Shah [20] studied the combination of financial news and price data for stock market forecasting with limited experimental validation and no user interaction or learning simulation component.

Taken together, the existing literature confirms a consistent pattern: systems that prioritise educational value tend to sacrifice trading realism, while systems that focus on technical trading capability rarely address the needs of learners. TradePro Elite is designed to resolve this tension by combining live market data, AI-driven insights, and community-based learning within a single risk-free platform.

III. SYSTEM DESIGN AND ARCHITECTURE

A. System Architecture

TradePro Elite is built on a modular, multi-layered architecture designed to handle real-time trading operations at scale. The React and TypeScript frontend provides a responsive interface for tracking live prices, placing trades, and managing portfolios, with WebSocket connections ensuring updates are delivered the moment prices change. The Node.js and Express.js backend manages authentication, trading logic, and transaction processing. User and trading data is stored in PostgreSQL via Prisma ORM, while Redis caching keeps latency low under high-frequency operations. An AI layer is integrated throughout the system, supporting price prediction, sentiment analysis, and portfolio optimisation.

B. Frontend Architecture

The frontend is developed using React 18 with TypeScript, bundled through Vite for fast build times and efficient hot module replacement during development. Tailwind CSS combined with shadcn/ui components provides the visual foundation, and the Chivo Mono Regular typeface is applied globally to reinforce the professional trading platform aesthetic. Zustand handles client-side state management, keeping component trees lean and avoiding unnecessary re-renders. Charts are rendered using a combination of Chart.js, Recharts, D3.js, and Lightweight Charts depending on the use case. Framer Motion handles transitions and animations. Socket.io Client maintains the WebSocket connection to the backend for live data delivery.

C. Backend Architecture

The backend runs on Node.js with TypeScript under an Express.js framework. It is responsible for authentication via Supabase Auth, core trading operations, portfolio management, and transaction processing. Socket.io manages the real-time event layer. Background jobs are handled through Bull Queue and Node Cron, covering tasks such as scheduled data refreshes and alert processing. Security is enforced through Helmet, CORS configuration, and rate limiting.

D. Data Layer

PostgreSQL serves as the primary database, with Prisma ORM managing schema migrations and query construction. The schema covers the following core models: User for authentication and profile data, Stock for Indian market securities with metadata, Portfolio for user holdings and positions, Transaction for the complete trading history, PriceHistory for chart data, AIInsight for AI-generated predictions and analysis,

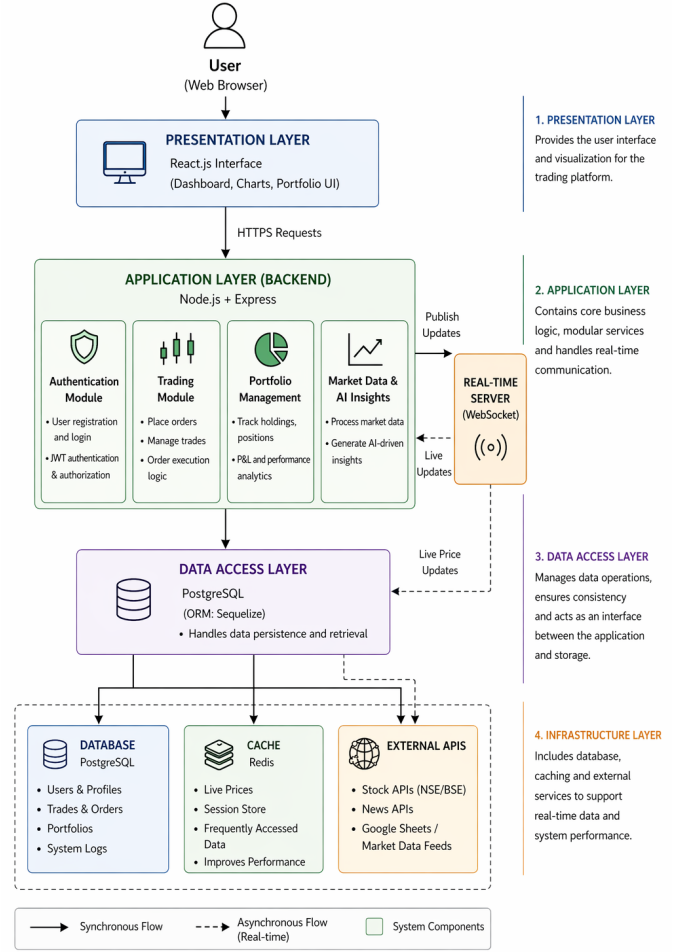


Fig. 1. TradePro Elite System Architecture

Alert for price notifications, Watchlist for user-defined stock lists, and UserAchievement for gamification and badges. Redis is used for caching frequently accessed data to reduce database load and improve response times.

E. Real-Time Market Data Pipeline

Live NSE and BSE stock prices are sourced through two complementary channels. Google Sheets acts as a live data relay, with market prices updated in real time and fetched by the backend through the Google Sheets API. The Fyers API provides professional-grade market data as a secondary and validation source. This dual-source approach improves reliability and reduces the risk of data gaps during high-activity market periods. The MarketDataService processes incoming price data and distributes updates to connected clients via WebSocket events.

IV. IMPLEMENTED FEATURES

A. Core Trading System

Each user account is initialised with a simulated balance of Rs.10,00,000. The trading system supports multiple order

types including market orders, limit orders, stop-loss, and take-profit. Users can buy and sell stocks across NSE and BSE-listed securities, with all transactions recorded in full. The portfolio view displays current holdings, average cost, current value, and real-time profit and loss calculations. A complete transaction history is available with filtering and analytics.

B. Market Data and Analytics

Live stock cards display real-time price updates with animated price change indicators. The platform tracks Nifty 50, Sensex, Bank Nifty, and over 25 additional market indices. Sector analysis covers more than 15 industry sectors with performance metrics and trend data. Interactive price charts support multiple timeframes. A market heatmap provides a visual overview of sector-level performance. Advanced search and filtering tools allow users to find and compare stocks efficiently.

C. AI-Powered Features

The AI dashboard serves as the central interface for machine learning-driven insights. The AIPredictionService generates price movement forecasts based on historical patterns and market signals. The SentimentAnalysisService processes financial news to produce sentiment scores for individual stocks. The PortfolioOptimizationService analyses a user's current holdings and suggests rebalancing actions to improve risk-adjusted returns. The RiskAssessmentService runs automated analysis of portfolio exposure and generates alerts when risk thresholds are approached. An interactive AI assistant is also available to answer trading-related queries and provide contextual advice.

D. Social Trading

The social trading module allows users to publish posts sharing trading insights, strategies, and market observations. Trading strategies can be formally published and followed by other users. Portfolio performance can be shared publicly to enable peer comparison and learning. A leaderboard ranks users by portfolio performance, driving healthy competition. A community feed aggregates posts and activity from across the platform.

E. Portfolio Analytics

The analytics module provides a comprehensive view of portfolio performance over time. Key metrics include Value at Risk (VaR), Sharpe ratio, and maximum drawdown. Sector allocation charts show how the portfolio is distributed across industries. Real-time profit and loss tracking is available at both the position and portfolio level. Performance charts display the progression of portfolio value over user-defined time windows.

F. User Experience

The platform is designed mobile-first and is fully responsive across device sizes. The dark theme uses a colour palette built around blue (#3B82F6) for primary actions, green (#10B981) for gains, red (#EF4444) for losses, and dark grays (#0F172A, #1E293B) for backgrounds. Glass morphism effects, subtle

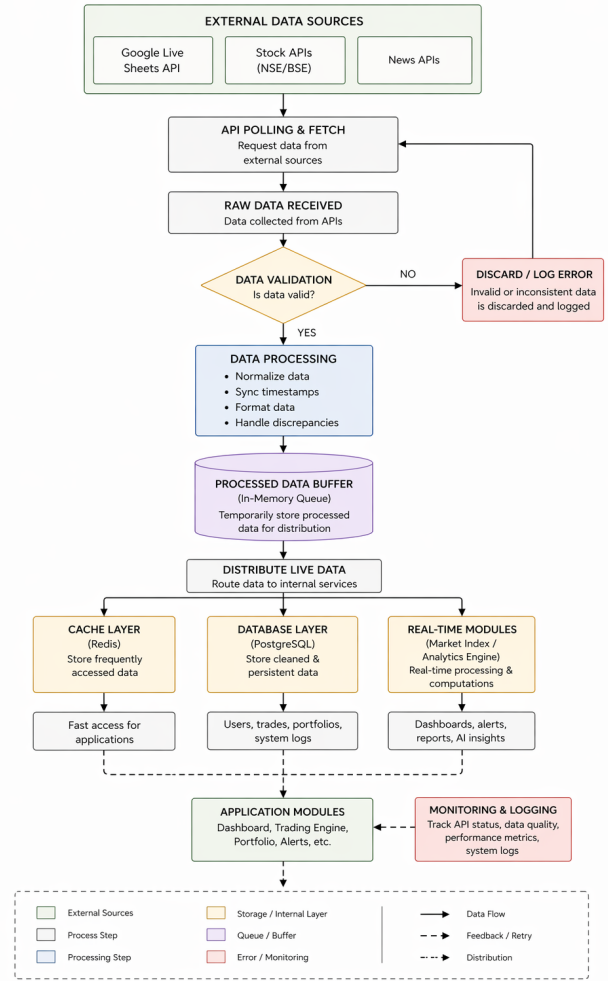


Fig. 2. Market Data Flow: From NSE/BSE Sources to User Interface

animations, and professional gradients contribute to the visual identity. Real-time price change notifications keep users informed without requiring manual refresh.

V. BACKEND SERVICES

A. Market Data Services

The MarketDataService is responsible for fetching live prices from Google Sheets and the Fyers API and distributing them to the rest of the system. The SectorAnalysisService aggregates stock-level data into sector-wide performance metrics. The SplitDetectionService monitors for stock split events and adjusts historical price records and holdings accordingly. The StockEventsDatabaseService tracks corporate actions including dividends, bonuses, and rights issues.

B. Trading Services

The TradingService handles all core trading operations, including order validation, execution, and confirmation. The PortfolioService maintains the state of each user's holdings and recalculates position values as prices update. The TransactionService records every trade with a full audit trail. The

WatchlistService manages user-defined watchlists and triggers alerts when watched stocks hit user-configured price levels.

VI. COMPARATIVE ANALYSIS

Table I compares TradePro Elite against representative existing platforms and studies across the key dimensions identified in the literature survey.

TABLE I
COMPARATIVE ANALYSIS OF TRADING SIMULATION PLATFORMS

Feature	Sarawade [1]	Bhattacharya [4]	Trivedi [9]	TradePro Elite
Live Market Data	No	No	No	Yes
AI Insights	No	No	No	Yes
Sentiment Analysis	No	No	No	Yes
Social Trading	No	No	No	Yes
Portfolio Analytics	Ltd	No	Ltd	Yes
Multiple Order Types	Ltd	No	No	Yes
Risk Metrics	No	No	No	Yes
Community Features	No	No	No	Yes
NSE/BSE Integration	No	No	No	Yes

A. Limitations

The platform currently depends on the availability and accuracy of the Google Sheets data relay and the Fyers API. Any disruption to either source affects the quality of live data delivered to users. The AI prediction models are based on historical price patterns and sentiment signals and do not account for macroeconomic shocks or events outside the training distribution. Options, futures, and derivatives trading are not yet implemented, which limits the platform's coverage of the full Indian market ecosystem. The social trading module currently lacks moderation tooling, which will become a concern at larger user scales. Remote or API-based trading integrations with actual brokerage accounts are outside the current scope.

VII. FUTURE ROADMAP

Development is planned across five phases. The first phase, targeting the next three months, focuses on enhanced AI capabilities including LSTM and Transformer models for price prediction, AI-powered chart pattern recognition, a real-time news feed with sentiment integration, and support for user-defined AI trading strategies.

The second phase, planned for three to six months out, introduces options and futures simulation with Greeks calculation, algorithmic trading with backtesting, and monthly paper trading competitions.

The third phase, spanning six to nine months, develops TradePro Elite into a full educational platform with structured trading courses, live webinars, proficiency certifications, and mentorship connections to professional traders.

The fourth phase, targeting nine to twelve months, adds enterprise features including institutional accounts, a public API for third-party integrations, a white-label solution for brokers, and native iOS and Android applications.

The fifth phase, beyond twelve months, extends the platform to international markets including US and European exchanges, cryptocurrency trading simulation, foreign exchange, and commodities.

VIII. CONCLUSION

TradePro Elite demonstrates that the gap between educational simulation and professional-grade trading tools does not need to exist. By connecting directly to live NSE and BSE market data through Google Sheets and the Fyers API, integrating an AI layer that covers prediction, sentiment analysis, and portfolio optimisation, and building community features that make learning social, the platform gives users an experience that is both authentic and risk-free.

The modular architecture ensures that each layer of the system can evolve independently as requirements grow. The AI services can be upgraded to more sophisticated models without touching the trading or data layers. The social module can scale with the user base without affecting core trading performance. The data pipeline can incorporate additional sources as they become available.

What makes TradePro Elite worth pursuing is not any single feature but the combination: real data, real tools, real market conditions, and no real money at risk. That combination is what serious trading education requires, and it is what the existing landscape has consistently failed to deliver.

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